

THANH D. PHAM

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RESEARCH INTERESTS

My current research focuses on developing principled approaches that enable AI systems to learn and adapt continuously and efficiently. I have a primary interest in **Knowledge Distillation**, **Neural Collapse**, **Continual Learning** and **DETR-based** model. I am also open to diversifying my research across various areas in the future.

My current research focuses on developing principled approaches for end-to-end visual recognition with transformer-based architectures. I have a primary interest in **DETR-based** model, **Interactive Segmentation** and **Remote Sensing Imagery**. I am also open to diversifying my research across various areas in the future.

EDUCATION

Bachelor of Data Science and Artificial Intelligence

2021 – 2025

Hanoi University of Science and Technology

High School for Gifted Students, Hanoi National University of Education

2018 – 2021

Mathematics Team Member; represented the school in regional and national mathematics competitions.

RESEARCH EXPERIENCE

AI Research Resident

Jul 2025 – Now

FPT Software AI Center

Hanoi, Vietnam

Advisor: Prof. Tan Pham

- Exploring Interactive Segmentation in Remote Sensing (ISRS) methods, and its applications. Our research investigates detection-guided interactive segmentation framework that leverages inter-object correlations to propagate user clicks across same-class objects.
- Developing a few-shot learning framework for diffusion-generated synthetic data. Our research introduces novel training paradigm that explicitly balances trade-off between fidelity and diversity when training with synthetic data.

Research Assistant

Dec 2022 – Sep 2025

Data Science Laboratory, BKAI, HUST

Hanoi, Vietnam

Advisor: Prof. Linh Ngo Van

- Worked in the Continual Learning research team. Our research improve prompt-based and prototype-based methods in Continual Learning by exploring its connection to Neural Collapse, leading to mitigating catastrophic forgetting and better performance.

Primary Researcher

2024

Class-Incremental Learning for Fine-Grained Object Detection (funded by Naver)

- Led a Naver-funded project on class-incremental learning strategies for fine-grained object detection.

PUBLICATIONS

1. **Thanh Duc Pham**, Nam Le Hai, Linh Ngo Van, Nguyen Thi Ngoc Diep, Sang Dinh, Thien Huu Nguyen. *Mitigating Non-Representative Prototypes and Representation Bias in Few-Shot Continual Relation Extraction*. Annual Meeting of the Association for Computational Linguistics (**ACL 2025**), Main Conference (**CCF A**).

PREPRINTS

1. **Thanh Duc Pham**, Anh Nguyen, Duong Duc Hieu, Minh-Tan Pham. *ISRS-DETR: Detection-Guided Click Propagation for Remote Sensing Interactive Segmentation*. **Under Review**.

2. **Thanh Duc Pham**, Lan-Cuong Nguyen, Dung D. Le, Minh-Tan Pham. *SyNC: Balancing Fidelity and Diversity of Synthetic Data Representations in CLIP-based Few-Shot Learning via Neural Collapse*. **Under Review**.
3. **Duc Thanh Pham**, Hong Dang Nguyen, Nhat Minh Nguyen Quoc, Linh Ngo Van, Sang Dinh Viet, Duc Anh Nguyen. *Hier-DETR: Hierarchical Neural Collapse Detection Transformer for Class Incremental Object Detection*. **Naver Project**.

AWARDS

Best Thesis Presentation Award
Hanoi University of Science and Technology (HUST)
Awarded for students with outstanding thesis presentation

2025

Silver Medal, National Student Mathematics Olympiad
Vietnam Mathematical Society (VMS)
National mathematics competition for undergraduate students in Vietnam

2022

VIASM Scholarship
Vietnam Institute for Advanced Study in Mathematics
Scholarship for students with exceptional achievement in mathematics

Vietnam Mathematical Olympiad (Encouragement Prize)
High School for Gifted Students, Hanoi National University of Education
The most prestigious math competition for high school students in Vietnam

2020

Gold Medal, Northern Mathematical Olympiad
High School for Gifted Students, Hanoi National University of Education
Regional mathematics olympiad for gifted high school students in northern Vietnam

2019

ACADEMIC SERVICES

Reviewer at AAAI 2025, ACL 2025, ICLR 2026 and ICML 2026

LANGUAGES

Vietnamese
Native

English
Professional working proficiency (IELTS 7.0)

SKILLS

Programming
Python

Technical
Math, Statistics, Machine Learning, Computer Vision

Libraries
PyTorch, Huggingface, NumPy, Pandas, Scikit-learn

REFERENCES

Prof. Minh-Tan Pham
IRISA, Université Bretagne Sud, France

Prof. Thien Huu Nguyen
University of Oregon, USA

Prof. Linh Ngo Van
School of Information and Communication Technology,
Hanoi University of Science and Technology, Vietnam